S4 Supplementary Clinical Summary

What the three empirical anchors do and do not mean at the bedside

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Short title: S4 Clinical - NDT 1.0

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Scope and intent

> **Important: no intervention, no biomarker, no consciousness meter**
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> The flagship is a theoretical framework. Companion analyses are observational. No treatment was tested. No individual-level prediction is licensed. Geometric summaries are not a CPC, a sleep-stage replacement, or a test of awareness.

This supplement is for clinically oriented readers. The goal is translation without amplification. Full methods and numbers live in the canonical companion articles (S2).

What was studied

Three constraints, three questions.

1. **N2 spindles** in healthy overnight recordings (ANPHY, BOAS, RichSleep). Question: does a discrete rhythm-defined event coincide with chart reconfiguration?
2. **Early EEG after cardiac arrest** (I-CARE, first 12 h after EEG-anchor). Question: does later good versus poor outcome separate more clearly in short-horizon futures and local flow than in static location?
3. **Simultaneous Neuropixels and surface EEG** (DANDI 000458). Question: does meso–macro association license a local operator?

Main findings, in order of clinical salience

1. After cardiac arrest, "how the record looks" is not the whole story

On the canonical invalidity-fixed merge (301 patients; 281 in CPC 1 versus CPC 5), short-horizon reachability and local deformation carry pooled information that static MNPS spread does not exhaust. Geometry added a bounded increment (AUROC $0.599 \rightarrow 0.662$) on a **feature-limited** clinical+qEEG baseline that omits SSEP, pupillometry, NSE, motor score, and imaging. Withdrawal-of-life-sustaining-therapy and sedation annotations were unavailable. Read as recovery-linked geometry, not viability and not prognostication.

2. Spindles move a chart coordinate; that is not therapy

A timing-locked rise in stratified d_n replicates on two datasets. It does not show that spindles clamp variance, write memory, or require a particular autonomic state. Matched non-spindle controls are incomplete.

3. Cross-scale contact is not a local law

Mesoscopic summaries can covary with EEG features under nulls. The tested EEG-only chart does not recover Jacobian or reachability operators. A correlation plus a pretty embedding is still Level 1.

What was not found

No universal "NDT signature" of unconsciousness, depression, psychosis, or recovery. Different conditions move different layers (main-text dissociation table). Reduced reachable volume is not itself reduced consciousness. Complexity measures can separate global state from within-state experience; NDT reachability is not licensed to do better until that test is run.

What clinicians should not conclude

- Do not treat MNPS, reachability, or Jacobian summaries as bedside scores.
- Do not equate CPC 5 with a geometric death state.
- Do not treat spindle-locked d_n as a target for stimulation or drugs.
- Do not read DANDI 000458 as evidence that EEG implements NDT operators.

What would increase clinical confidence

Independent implementation of the same contracts; cleaner endpoints (including WLST where ethically available); within-state experience sampling against accessibility geometry; residual-quality gates before any operator claim.

Summary for clinical readers

NDT asks what a neural state can still become. Current clinical companions constrain that question in sleep, post-arrest EEG, and a meso–macro bridge. They do not replace clinical judgment, multimodal prognostication, or a theory of experience.

Bibliography